Towards an Integrated and Realtime Wayfinding Framework for Flood Events

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ABSTRACT

City planners can encounter severe challenges during natural disasters. Flooding, for example, is considered as the number one cause of infrastructure damage across the world. Flooding can have a significant impact on personal property, commercial assets, and essential infrastructure, including power and gas delivery, and transportation. This paper focuses on the effects of flooding on transportation systems in the State of Iowa under a variety of flood scenarios, including 50, 100, 200, and 500-year return probabilities. We explore the impacts of flooding from institutional support (e.g., services including police, fire, and EMS) and general population (i.e., individuals distributed across cities) perspectives. Graph-theoretic methods are used in this study to determine the effects of flooding on road networks due to potential removal of paths that were once viable. This paper presents preliminary research into flood impacts on road infrastructure in Iowa and the development of an integrated real-time framework for analyzing those impacts. In future work, we plan to extend the framework developed in this study to provide a generalized decision-support system for cities and individuals. The framework will be open so city planners will be able explore "what if" flooding scenarios to find vulnerable areas and populations in their jurisdiction. These areas can be made more resilient to flooding effects by increasing the elevation of important roads, changing flow patterns, or increasing the height of bridges.

CCS CONCEPTS

• **Information systems → Temporal data**; **Location based services**; *Network data models*; *Data streaming*.

KEYWORDS

Wayfinding, flooding, real-time analytics, flood resilience

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1 INTRODUCTION

Real-time and integrated information frameworks [10, 26] with crowdsourced [24] and structured data integration [6, 22], knowledge generation [21], information visualization [20], data analytics [27, 28], modeling and forecasting [23], and decision-support [5, 7] capabilities have been studied extensively in recent years for flood risk management and environmental planning for natural disasters.

Rainfall patterns have changed over the past 50 years causing an increase in the frequency and impact of natural disasters such as flooding [16]. Urban development contributes to increased surface flow and decreased infiltration. Flooding affects most of the world with many locations experiencing annual occurrences. Due to natural disasters and human-made causes, billions of dollars and hundreds of human lives can be lost each year. Between 1984 and 2014, the Federal Emergency Management Agency (FEMA) has spent an estimated $48.6 billion to recover public infrastructure (ex. roads) from flood damages. Annually, more than $2 billion is paid by the National Flood Insurance Program (NFIP) for flood-related disaster [8]. The State of Iowa has suffered multiple devastating flood events, over the last 20 years, leading to severe losses. In 2008, the total estimated damage across the State was $10 billion [30].

During a flood event, it is challenging for people to continue daily routines like gathering supplies, getting groceries, and going to school or work. There is an increased risk of injury or death when using transportation networks in flood conditions due to changing environmental conditions and a lack of precision when assessing hazardous conditions in advance. If a roadway becomes inundated by floodwaters, it is essential to identify these road segments and provide alternative routes. Managing transportation networks under dynamic disaster conditions is critical for flood disaster management.

2 BACKGROUND

Flooding can severely impact infrastructure in rural and urban settings. Flooding can result from heavy rain, river overflow, hurricanes, dam collapse, and snow melts. Moreover, it can cause damage to different types of structures, such as buildings, roads, and vehicles [25]. During a flood event, decisions about evacuation routes and the best locations for shelters are paramount.

There are general trends in research on wayfinding and navigation during disasters. Some papers have developed optimal evacuation strategies such as determining evacuation routes [4, 9], setting a logistic time to map out the evacuation places from the flood area [1], minimizing the expected total evacuation time [29], and the optimal placement of emergency shelters [14]. In addition, other researchers have studied routing for emergency services during

a disaster [11, 15] and routing supply vehicles for deliveries of essential goods and services [3, 19].

It is essential to evaluate the resilience of road structures, especially road segments and intersections, for different types of disasters. Not only are there concerns about physical safety for traversing roads, it is necessary to understand roads as networks. They provide a valuable infrastructure that can change in unexpected ways during disasters. For example, routes may become altered or non-existent dynamically. Researchers have studied the adverse impacts of a flood event on transportation infrastructure [12] and the probability of the network blocking during single node flooding [17]. During a disaster, it is a challenge to plan for evacuation routes since the number of evacuees often exceed the capacity of a route. Kim, George, Shekhar [13] propose heuristics to large networks to overcome capacity constraints of a network. We seek to extend previous research by creating an open and dynamic framework that can be used to expose and visualize infrastructure weakness, be used in âĂIJwhat ifâĂİ scenarios and simulations, and for urban and regional planning.

3 METHODS

In this research, we use data from the OpenStreetMap (OSM) project to build a dual-representation of road networks that allow exploration of the network system using spatial analyses and graph-theoretic methods. The first representation is a spatially and topologically explicit network model used in a geographical information system (GIS). The second representation is a topologically-explicit graph representation used by graph-theoretic methods to explore connectedness and adjacency of graph features (i.e., nodes and edges). We also use Light Detection and Ranging data (LIDAR) to calculate bridge deck heights, a bare-earth digital elevation model (DEM) for base elevations of the terrain, and flood depth models to identify bridge closures from spatial information that is publicly available. Bridge inventory data from the National Bridge Inventory (NBI) database is used to find the locations of bridges and culverts on both representations of the road network.

We encountered many issues with the spatial data used in this research. Some of the road network data has missing or incomplete regions, and those data are of varying quality. However, for this research, it is important to use data that is freely available. We also experienced issues with data aligned to different spatial extents, including the flood depth model (HUC8 watersheds), the DEM (arbitrary tessellation), and 2x2 km blocks for the LIDAR data. This required significant data processing to align these datasets into HUC8 watershed extents. Another significant issue with the data was that the NBI dataset does not contain the deck height of bridges. To determine whether a bridge is closed under various flood scenarios, our system uses bridge point locations from the NBI, the DEM, lidar-derived elevation models, and flood depth models created at the Iowa Flood Center. NBI data provides a single point defining the start or end of the bridge so we buffer the bridge point by half of the bridge length to produce 2+ points when intersected with the road network, one of which could be the approximate bridge midpoint. Our assumption is that the bridge deck height will be the highest elevation on the lidar-derived DEM and the lowest elevation on the bare-earth DEM. Flood depth will likely be highest near the bridge midpoint as well. Once the midpoint has been selected, we compare the deck height to the bare-earth DEM plus the flood model depth to see the difference between those elevations and consider the bridge closed if the deck is overtopped.

We use many graph-theoretic methods to determine the initial topological structure of the network graph. We calculate baseline statistics on the connectedness of nodes and edges within the graph. Under varying flood probabilities, edges of the graph that are inundated will be removed so the underlying topology may change. These changes will increase driving times or distances for some locations within the study area and may result in the creation of "island" or isolated populations that cannot access essential services. We also use all-pairs and single-source shortest path analyses, betweenness centrality [2], closeness centrality, and edge betweenness centrality measures to understand the topological structure of the road network under various flood probabilities.

We use a Python library [18] for graph operations in this research which uses the betweenness centrality measures created by Brandes [2]. Although betweenness centrality provides a basic understanding on the network topology, it isn't adequate to address the dynamics of flooding impacts. We use it mostly to understand how the removal of network edges affects the subgraph topology. Once edges are removed from the graph due to inundation, we also run k-shortest path analyses to understand aggregate effects of those changes. For example, to better understand the real effects of the removal of edges, we look from an institutional support perspective and a general population perspective. An institutional support perspective examines accessibility from and to service providers such as hospitals, emergency medical services, and police services.

Accessibility is measured in terms of an increase in travel time or distance, and whether these services can be reached at all. In our initial analyses, we have seen cases where populations can become isolated and cut off from essential services under extreme flood scenarios. Aggregate statistics on these measures show increases in travel time and distance based solely on changes in the topological structure of the network. We also examine shortest paths from network nodes to amenities such as grocery stores, schools, and churches. This provides a basic understanding of the difficulty of reaching certain locations for all other locations. For example, will there be an increase in the distance or time required to reach the closest grocery store? At this time, however, these measures are based on the topological structure of the network and not the actual distribution of population or individual set choices.

4 RESULTS AND DISCUSSION

As expected, we see changes in the network topology under various flood scenarios. Figures 2 and 3 show Webster City, Iowa; a small city north of Des Moines. We selected this city for this paper because it is situated at the confluence of a river and several creeks.

The inset map shows preliminary results from betweenness centrality measures on the baseline topology (no flood) and a 50-year flood scenario. The inset shows the betweenness centrality measures for each scenario using graduated symbols. Larger circles indicate nodes with greater importance based on betweenness centrality measures, i.e., more shortest paths pass through them. If a node decreases in importance between a no-flood scenario and under flooding, then the viewer will see a green outline around the node symbol. In both maps, flood scenario nodes (in purple)

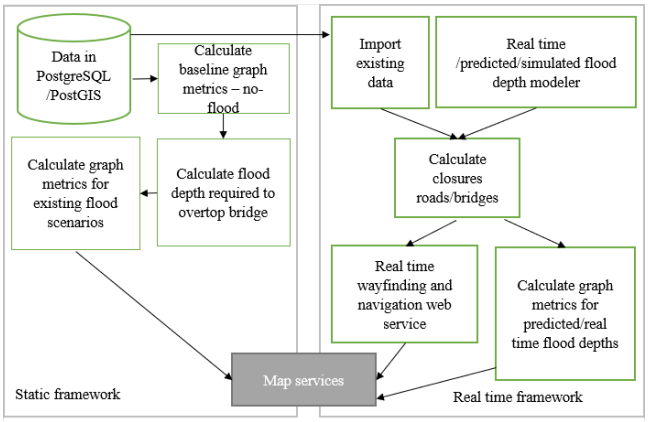

Figure 1: Process Diagram

that gain prominence will obscure the baseline nodes. Red edges in the inset show road segments that are closed due to flooding. Since these edges are removed from the graph used to calculate the centrality measures, some nodes may increase or decrease in the number of shortest paths that pass through those nodes. Betweenness centrality is commonly used to explore the relative importance of nodes due to the number of shortest paths that pass through them. Figure 2 shows the change in the importance of nodes using betweenness centrality measures in Webster City, Iowa under baseline conditions and during a 50-year flood extent. Figure 3 shows the same extent under a 500-year flood probability.

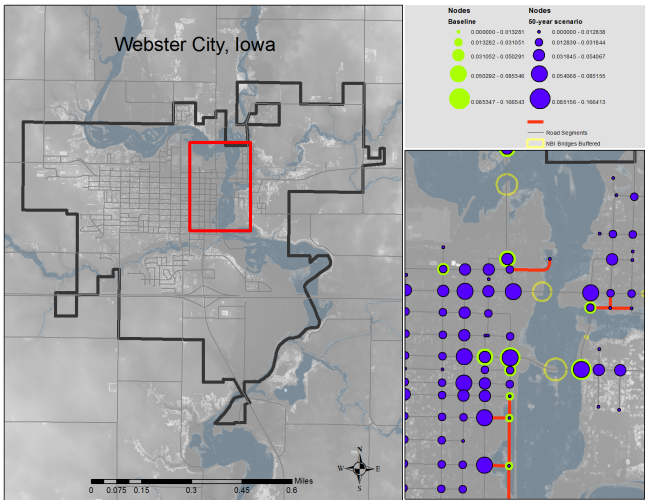

Figure 2: Webster City, Iowa 50-year flood scenario

5 CONCLUSIONS AND FUTURE WORK

This work focuses on the development of an integrated and real-time framework for analyzing the potential impacts of flooding on road networks across Iowa. As expected, we find differential flooding impacts on these networks based on the unique environmental settings for those locations and the flood probability model used in the analysis. We used traditional graph metrics to understand the topological structure of road networks of cities in Iowa and to demonstrate how changing conditions alter those structures. In further research, we will extend those metrics to account for the distribution of population, households, and vehicles within a city. This is expected to provide a more insightful understanding of the relative importance of edges and nodes within the road network.

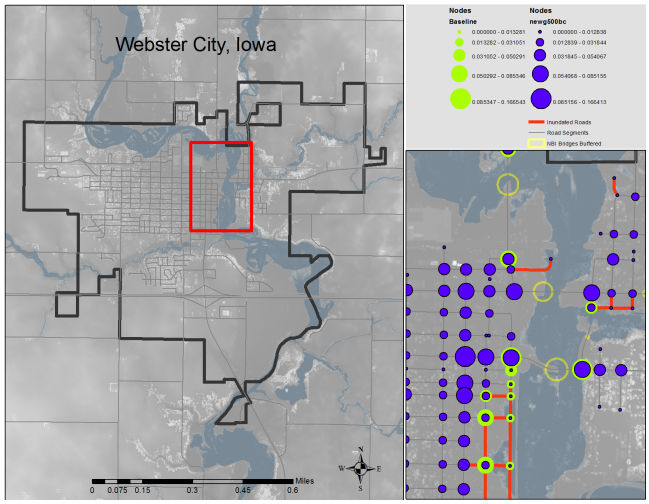

Figure 3: Webster City, Iowa 500-year flood scenario

We will also couple historic, real-time, and predicted flood levels against the road network infrastructure to explore "what if" scenarios. City planners, for example, could use our framework to explore situations that may not have occurred previously. Those results could be used to indicate potential infrastructure modifications that could make cities more resilient to flood events. In a related study, we have developed an on-demand real-time flood inundation service using Lidar datasets. That service creates flood extent and depth models for a location using predicted flood heights. If these two frameworks are coupled together, then they could be used as a decision-support system for states, regions, counties, cities, or individuals. A decision-support system like that could help minimize the loss of life and infrastructure damage in flooding events. For example, individuals could submit their location (e.g. house or campsite) and the combined systems would calculate network accessibility under those conditions. This could provide valuable decision-support regarding the potential impacts of nearby flooding. This could help mitigate some of the uncertainty, and therefore risk, that is commonly seen in flood events. Often people do not understand how their current location will be affected by predicted flood levels so they can make the unwise decision to remain in that location and become isolated.

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REFERENCES

[1] M. Borowska-Stefańska, S. Wiśniewski, and M. T. Andrei. 2017. Accessibility to places of evacuation for inhabitants of flood-prone areas in Mazovia Province. *Geomatics and Environmental Engineering* 11 (2017).

[2] U. Brandes. 2001. A Faster Algorithm for Betweenness Centrality. *Journal of Mathematical Sociology* 25, 2 (2001), 163–177.

[3] A. M. Campbell, D. Vandenbussche, and W. Hermann. 2008. Routing for relief efforts. *Transportation Science* 42, 2 (2008), 127–145.

[4] V. Campos, R. Bandeira, and A. Bandeira. 2012. A method for evacuation route planning in disaster situations. *Procedia-Social and Behavioral Sciences* 54 (2012), 503–512.

[5] A. Carson, M. Windsor, H. Hill, T. Haigh, N. Wall, J. Smith, and M. Muste. 2018. Serious gaming for participatory planning of multi-hazard mitigation. *International Journal of River Basin Management* 16, 3 (2018), 379–391.

[6] I. Demir and R. Szczepanek. 2017. Optimization of river network representation data models for web-based systems. *Earth and Space Science* 4, 6 (2017), 336–347.

[7] I. Demir, E. Yildirim, Y. Sermet, and M. A. Sit. 2018. FLOODSS: Iowa flood information system as a generalized flood cyberinfrastructure. *International Journal of River Basin Management* 16, 3 (2018), 393–400.

[8] M. Denchak. 2019. Flooding and Climate Change: Everything You Need to Know. https://www.nrdc.org/stories/flooding-and-climate-change-everything-you-need-know/.

[9] M. Goerigk, K. Deghdak, and P. HeÃŞler. 2014. A comprehensive evacuation planning model and genetic solution algorithm . *Transportation research part E: logistics and transportation review* 71 (2014), 82–97.

[10] C. S. Jones, C. A. Davis, C. W. Drake, K. E. Schilling, S. H. Debionne, D. W. Gilles, and L. J. Weber. 2018. Iowa statewide stream nitrate load calculated using in situ sensor network. *JAWRA Journal of the American Water Resources Association* 54, 2 (2018), 471–486.

[11] A. Jotshi, Q. Gong, and R. Batta. 2009. Dispatching and routing of emergency vehicles in disaster mitigation using data fusion. *Socio-Economic Planning Sciences* 43, 1 (2009), 1–24.

[12] A. Kermanshah and S. Derrible. 2017. Robustness of road systems to extreme flooding: using elements of GIS, travel demand, and network science. *Natural hazards* 86, 1 (2017), 151–164.

[13] S. Kim, B. George, and S. Shekhar. 2007. Evacuation route planning: Scalable heuristics. *Proceedings of the 15th annual ACM international symposium on Advances in geographic information systems*.

[14] S. Kongsomsaksakul, C. Yang, and A. Chen. 2005. Journal of the Eastern Asia Society for Transportation Studies. *Shelter location-allocation model for flood evacuation planning* 6 (2005), 4237–4252.

[15] V. Mali, M. Rao, and S. S. Mantha. 2012. Enhanced routing in disaster management based on GIS. *International Conference on Intuitive Systems Solutions*, 5–6.

[16] I. Mallakpour and G. Villarini. 2016. Investigating the relationship between the frequency of flooding over the central United States and large-scale climate. *Advances in Water Resources* 92 (2016), 159–171.

[17] Z. H. Nasralla, T. E. El-Gorashi, M. O. Musa, and J. M. Elmirghani. 2016. Routing post-disaster traffic floods in optical core networks. *2016 International Conference on Optical Network Design and Modeling (ONDM)*, 1–5.

[18] Networkx. 2019. https://networkx.github.io/.

[19] L. Özdamar and O. Demir. 2012. A hierarchical clustering and routing procedure for large scale disaster relief logistics planning. *Transportation Research Part E: Logistics and Transportation Review* 48, 3 (2012), 591–602.

[20] B. C. Seo, M. Keem, R. Hammond, I. Demir, and W. F. Krajewski. 2019. A pilot infrastructure for searching rainfall metadata and generating rainfall product using the big data of NEXRAD. *Environmental Modelling Software* 117 (2019), 69–75.

[21] Y. Sermet and I. Demir. 2018. An intelligent system on knowledge generation and communication about flooding . *Environmental Modelling Software* 108 (2018), 51–60.

[22] Y. Sermet and I. Demir. 2019. Towards an information centric flood ontology for information management and communication. *Earth Science Informatics* (2019), 1–11.

[23] M. Sit and I. Demir. 2019. Decentralized flood forecasting using deep neural networks. *arXiv preprint arXiv:1902.02308* (2019).

[24] M. A. Sit, C. Koylu, and I. Demir. 2019. Identifying disaster-related tweets and their semantic, spatial and temporal context using deep learning, natural language processing and spatial analysis: a case study of Hurricane Irma . *International Journal of Digital Earth* (2019), 1–25.

[25] E. Vlachos. 1995. Socio-economic impacts and consequences of extreme floods. *Challenges in Water Resources Management* 19 (1995).

[26] L. J. Weber, M. Muste, A. A. Bradley, A. A. Amado, I. Demir, C. W. Drake, and N. W. Thomas. 2018. The Iowa Watersheds Project: Iowa's prototype for engaging communities and professionals in watershed hazard mitigation. *International Journal of River Basin Management* 16, 3 (2018), 315–328.

[27] H. Xu, I. Demir, C. Koylu, and M. Muste. 2019. A web-based geovisual analytics platform for identifying potential contributors to culvert sedimentation. *Science of The Total Environment* 692 (2019), 806–817.

[28] E. Yildirim and I. Demir. 2019. An integrated web framework for HAZUS-MH flood loss estimation analysis. *Natural Hazards* (2019), 1–12.

[29] X. Zhao, Z. Y. Feng, Y. Li, and A. Bernard. 2016. Evacuation network optimization model with lane-based reversal and routing. *Mathematical Problems in Engineering* (2016).

[30] J. Zogg. 2014. *The Top Five Iowa Floods*. Technical Report. National Weather Service WFO: Des Moines, IA, USA.